

NZMPTA Milking Machine Test Summary

Farmers Name: M & L. Slaughter
Test Date: 20-3-26 Tester Name/Company: Greg Lister
Tester: Reg No. 1020 Tester Contact Details: 021 437 531

1. Key Performance Measurements

- a) Reserve ☒
b) Vacuum@ Receiver ☒
c) Regulation ☒
Pulsation ☒
Ratio 58-60%
Rate 60
D Phase > 30%

2. Safety

- a) Moving Parts ☒
e.g. guards on releasers, vacuum pumps, slide pulsators, all other shafts and belts etc
b) Stationary Parts ☒
c) Other (electrical, oil) ☒

Note: This Machine Test may identify hazards, however, it in no way guarantees safety compliance for any hazard. It is the farm owner's responsibility to ensure that all moving parts comply with WorkSafe and relevant NZ Safety Standard/s.

3. Condition of Components

- a) Vacuum Pump(s) and Releaser(s) ☒
b) Vacuum & Milk System ☒
c) Ancillary Equipment ☒
d) Clusters ☒
e) Regulator(s) ☒
f) Farm Gauge ☒
g) Pulsators ☒

4. Improvements

Rating

1/ Vacuum at receiver was higher than normal on arrival due to a new vacuum pump. Installed a filtered air bleed to balance vacuum. All good now (don't turn off)

min

3/ Vacuum Pump - Manufacturer recommends minimum speed of no less than 1000 rpm. Currently, yours is 700 rpm. Performance is slightly under minimum performance level - not an issue as you have eodles of reserve vacuum.

min.

min

Milk pump shaft coupling very worn - replace soon.

min.

3/ Clusters - 2x lines not pulled through to tension ring - done today
Lots of holed long milk & long pulse tubes - cut back today

min

min

3/ Regulator is missing the top cover/feltor - replace

min

Changed oil in milk pump gearbox today

Rating	Reason	A very clean & tidy shed		
Critical	Negatively affects milk quality	Negatively affects animal health/welfare	Serious risk to milker health and safety	
Major	Affects animal comfort	May cause a breakdown	Affects ability to harvest milk efficiently	Compromises milker safety
Minor	Potential to impact on milk quality	Potential to impact on animal health/welfare	Potential to impact on milker safety in the future	

Milking Machine Test Report Results

Activity	Standard Reference	Calculated Standard	Reading	✓ ✗						
Working Vacuum (Receiver)	≤ 50 kPa	42-46	42 kPa	✓						
Vacuum Regulation Deviation (1a - 1b)	+/- 2 kPa		0 kPa	✓						
Vacuum at Minimum Vacuum Pump Speed (1a - 1f)	< 2 kPa		1 kPa	✓						
Effective Reserve Or Cleaning Reserve	Compare Reading to greater of Standard Effective Reserve Or Standard Cleaning Reserve	950 1991	2770 lpm	✓						
Regulation Loss (2c - 2a)	≤ 35 l/min or 10% of manual reserve	≤ 277	0 lpm	✓						
Regulator Leakage (2e - 2b)	≤ 35 l/min or 5% of manual reserve	≤ 139	0 lpm	✓						
Vacuum Drop Between Regulator and Receiver (4b - 4a)	≤ 1 kPa		0.3 kPa	✓						
Vacuum Drop Between Pump(s) and Receiver (4d - 4a)	≤ 3 kPa		0.8 kPa	✓						
Regulation Sensitivity (5a - 1a)	≤ 1 kPa		+1 kPa	✓						
Plant Gauge Accuracy (differences between Farm Gauge & Test Gauge)	≤ 1 kPa Record largest deviation		0 kPa	✓						
Vacuum System Leakage (Airflow start - 10a)	≤ 5% of Vacuum Pump Capacity @ w/v	≤ 215	80 lpm	✓						
Milking System Leakage (10a - 10c)	≤ 10 lpm + 2 lpm per cluster	≤ 76	10 lpm	✓						
Automatic Cluster Remover (10c - 11a)	See Standards Manual		— lpm							
Cluster Air admission (10c - 12a)	≥ 4 l/min and ≤ 12 l/min Or < 35 l/min/cluster (vented liners)	140-420	190 lpm	✓						
Milk System Ancillary Equipment Leaks (12a - 14a)	See Standards Manual		— lpm							
Pulsator Consumption (14a - 14c)	35 l/cluster	≤ 1155	1000 lpm	✓						
Vacuum System Ancillary Equipment Leaks (14c - 14e)	See Standards Manual	≤ 30	10 lpm	✓						
Pulsator Airline Drop (1a - 15a)	≤ 2 kPa		10.9 kPa	✓						
Other Readings:										
Vacuum Pump Exhaust Pressure (9a)	See Standards Manual									
Effective Reserve (Vacuum off Cluster) (6b)	Guideline Only									
		Fall Off Vacuum Drop (3a - 3c)	Undershoot (3c - 3b)	Overshoot (3d - 3e)						
Regulation Characteristics	≤ 2 kPa	0.1 kPa	1.5 kPa	1.2 kPa						
Vacuum Pump(s)										
	Capacity at 50kPa	Standard Capacity	✓/✗	RPM Minimum	RPM Maximum	✓/✗	kW	✓/✗	Oil/Water Flow Rate	✓/✗
Pump 1	3660	≥ 3700	✗	700	1470		15	✓	6	✓
Pump 2				✗						
Pump 3										
Pump 4										

NZMPTA Milking Machine Test Record



Test No: 4596 Date: 20-3-26
 Tester (name): Greg Reg No: 1020 Expiry Date: 30-4-27 Telephone: 021 437 531
 Calibration Expiry Dates: Air Flow Meters 5-26 Pulsator Testers 5-26 Vacuum Gauges 5-26

Farm Owner: LM Farms Telephone: 021 722 960 M
 Occupier: Michael & Lenay Slater Telephone: 021 723 941 L
 Farm Location: West Coast Road, Te Kapa Dairy Factory & Supply No: FON 18108
 Postal Address: RD1

Type & Size: 33HLH13 (1.55M) Regulator Type: CSL Varivac Vacuum Pump: WRC 4
 Vacuum Pump Motor Size: 15kW No. & Type of Pulsator: 33xP100S-2xM Pulsation is 4x0 / 2x2
 Releaser Type & Motor Size: CSL Lohr 2.2kW w VSD Claw: AIK Shell: Wairarapa L
 Vented Liner: Yes No Milkline Size: 100mm Machine is Variable Speed Drive: Yes No Flushing Pulsation System: Yes No
 Herd Size: Last BMCC:

1 Vacuum Levels	
1a Working Vacuum (Receiver)	<u>42 / 45.2</u> kPa ☒
1b Nominal Working Vacuum	<u>42-46</u> kPa ☒
1c Vacuum Regulation Deviation (1a - 1b)	<u>0</u> kPa ☒
1d Working Vacuum (Regulator)	<u>42 / 45.3</u> kPa ☒
1e Working Vacuum (Vacuum Pump)	<u>42.2 / 45.7</u> kPa ☒
1f Working Vacuum @ minimum VP speed	<u>6.1</u> kPa ☒
2 Reserve Characteristics	
2a Effective Reserve	<u>31.10</u> l/min ☒
2b Air Flow 2 kPa below 1d	<u>31.10</u> l/min ☒
2c Manual Reserve	<u>31.10</u> l/min ☒
2d Regulation Loss (2c - 2a)	<u>0</u> l/min ☒
2e Air Flow 2 kPa below 1d	<u>31.10</u> l/min ☒
2f Regulator Leakage (2e - 2b)	<u>0.8</u> l/min ☒
2g Required Standard Effective Reserve	<u>9.8</u> l/min ☒
2h Required Cleaning Reserve	<u>19.91</u> l/min ☒
3 Regulation Characteristics	
3a Average Receiver Vacuum	<u>42.1</u> kPa ☒
3b Minimum Vacuum during Air Inlet	<u>40.5</u> kPa ☒
3c Average Vacuum during Air Inlet	<u>42.1</u> kPa ☒
3d Maximum Vacuum Increase	<u>43.1</u> kPa ☒
3e Average Vacuum after Stop of Air Inlet	<u>42.5</u> kPa ☒
3f Fall off Vacuum Drop (3a - 3c)	<u>0.2</u> kPa ☒
3g Regulation Undershoot (3c - 3b)	<u>0.3</u> kPa ☒
3h Regulation Overshoot (3d - 3e)	<u>1.2</u> kPa ☒
4 Airline Vacuum Drop	
4a Vacuum at Receiver	<u>39.7</u> kPa ☒
4b Vacuum at Regulator	<u>40</u> kPa ☒
4c Vacuum Drop Regulator/Receiver (4b - 4a)	<u>0.3</u> kPa ☒
4d Vacuum near Vacuum Pump(s)	<u>40.5</u> kPa ☒
4e Vacuum Drop Vacuum Pump/Receiver (4d - 4a)	<u>0.8</u> kPa ☒
5 Regulation Sensitivity	
5a Working Vacuum in Milking System	<u>48 / 43</u> kPa ☒
5b Regulation Sensitivity (5a - 1a)	<u>16</u> kPa ☒
6 Reserve (Vacuum off Cluster)	
6a Vacuum at Interceptor (Vacuum off Cluster)	<u>1.1</u> kPa ☒
6b Effective Reserve (Vacuum off Cluster)	<u>1.1</u> l/min ☒
7 Vacuum Gauge Accuracy	
Adjust AFM to 5 kPa below vacuum 4a	
7a Farm Gauge Vacuum	<u>35</u> kPa ☒
7b Test Gauge Vacuum	<u>35</u> kPa ☒
Gauge Error at -5kPa (7a - 7b)	<u>0</u> kPa ☒
7c Adjust AFM to Vacuum 4a	
7d Farm Gauge Vacuum	<u>40</u> kPa ☒
7e Test Gauge Vacuum	<u>40</u> kPa ☒
7f Gauge Error at Vacuum 4a (7d - 7e)	<u>0</u> kPa ☒
Adjust AFM to Vacuum 4a plus 5 kPa	
7g Farm Gauge Vacuum	<u>45</u> kPa ☒
7h Test Gauge Vacuum kPa	<u>45</u> kPa ☒
7i Gauge Error at +5 kPa (7g - 7h)	<u>0</u> kPa ☒

8 Vacuum Pump Capacity and Speed	
8a Pump Capacity	
50 kPa	<u>3460</u> l/min ☒
Pump 1	<u>3460</u> l/min ☒
Pump 2	<u>3460</u> l/min ☒
Pump 3	<u>3460</u> l/min ☒
Pump 4	<u>3460</u> l/min ☒
8b Minimum Pump Speed	<u>1150</u> rpm ☒
8c Maximum Pump Speed	<u>1150</u> rpm ☒
9 Vacuum Pump Exhaust Pressure	
Vacuum Pump(s) is not specification. Otherwise go to Air Flow Start	
9a Vacuum Pump Exhaust Pressure	<u>4300</u> l/min ☒
Air Flow Start	
Adjust AFM Vacuum 1e	
Pump Capacity at Working Vacuum	<u>4300</u> l/min ☒
10 Airline and Milk System Leakage	
10a Air Flow with Vacuum System Connected	<u>4220</u> l/min ☒
10b Vacuum System Leakage (Air Flow Start Minus 10a)	<u>80</u> l/min ☒
10c Connect Milk System (ancillary equipment disconnected)	<u>76</u> l/min ☒
10d Milking System Leakage (10a - 10c)	<u>4210</u> l/min ☒
11 Automatic Cluster Removers	
11a Air Flow ACR's Connected	<u>140-420</u> l/min ☒
11b ACR Consumption (10c - 11a)	<u>190</u> l/min ☒
12 Cluster Air Admission	
12a Connect Cluster Air Admission	<u>140-420</u> l/min ☒
12b Cluster Air Admission (10c - 12a)	<u>190</u> l/min ☒
13 Individual Cluster Air (see additional tests)	
14 Pulsators and Ancillary Equipment	
14a Air Flow with Milk System Ancillary Equipment Attached	<u>3020</u> l/min ☒
14b Milk System Ancillary Equipment Leakage (12a - 14a)	<u>10</u> l/min ☒
14c Airflow with Pulsators Connected	<u>3020</u> l/min ☒
14d Pulsator Consumption (14a - 14c)	<u>1000</u> l/min ☒
14e Airflow with Vacuum System Ancillary Equipment Connected	<u>3010</u> l/min ☒
14f Vacuum System Ancillary Equipment Leakage (14c - 14e)	<u>10</u> l/min ☒
15 Test Pulsation - Go to Pulsation Recording Sheet	
15a Lowest Maximum Pulsation Chamber Vacuum	<u>42.9</u> kPa ☒
15b Pulsator Airline Drop (1a minus 15a)	<u>10.9</u> kPa ☒
16 Final Check Tests	
16a Working Vacuum at Receiver (Compare to reading 1a)	<u>42</u> kPa ☒
16b Effective Reserve (Compare to reading 2a)	<u>2770</u> l/min ☒

Note: A full milking machine test must be completed at least annually only by a NZMPTA Registered Milking Machine Tester.

Milking Machine Additional Testing

(Optional Tests)

Activity	Reading	Standard	✓ X
Gate cylinder consumption:	lpm		
Milk Meter:	lpm		
Teat Spray:	10 lpm	≤ 30	✓
Other:	lpm		
Other:	lpm		
Other:	lpm		
Other:	lpm		
Vacuum Pump Oil/Water Flow Rate:	6	3.8-7	✓
Atmospheric Pressure:			
Leakage in Releaser:			
Bubbles in discharge line:			
Water drawn back:			
Bubbles in Receiver:	N/A	N/A	✓
Diaphragm Pump:			
Speed:	strokes/min		
Head One:	kPa	≥ 85	
Head Two:	kPa		
Head Three:	kPa		
Head Four:	kPa		
Head Five:	kPa		
Head Six:	kPa		
All Heads		≥ 85	
Cluster Fall-off air inlet:	lpm		
Peak Regulator Load/Safety Valve Test			
Transfer the appropriate standards from the Calculated Standards column of the Test Report Results form			
Effective Reserve or Cleaning Reserve	1991		
Vacuum System Leakage	215		
Milking system leakage	76		
Cluster Air Admission	420		
Milk System Ancillary Leakage	—		
Pulsator Consumption	1155		
Vacuum System Ancillary Leakage	30		
Regulation Leakage	139		
ACR Consumption	—		
Standard Pump Capacity Total	4026 lpm	Quote this total when recommending a new Vacuum Pump. Refer to module and Reserve Flowchart in standards book.	